



**Narnarayan Shastri Institute of Technology**  
**Institute of Forensic Sciences and Cyber Security, Jetalpur, Ahmedabad,**  
**Gujarat, India.**

**In Association with ECLearnix EdTech Pvt. Ltd., (Coimbatore, India & USA)**

**International Level Hackathon**  
**ECLearnix 360° 4.O Innovation Challenge**  
**Round 2 (Only for Round 1 shortlisted Team)**

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Organization  
Logo**

**Chosen Hackathon Domain :**

- Domain 1 - Open challenge Solutions (Project,case study etc)
- Domain 2 - Business Solutions
- Domain 3 - Edtech Solutions
- Domain 4 - IT solutions
- Domain 5 - Cyber Security & Forensic Science
- Domain 6 - Healthcare & Allied

**Team Details :**

- Team Name : CODERX
- Domain Name : open challenge solutions
- Single Point of Contact (SPOC): NITHIN PRANAV T
- SPOC Email ID : nithinpranav010@gmail.com
- SPOC Mobile Number :9500190447

# TEAM MEMBER'S NAMES & CONTACT DETAILS

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Organization  
Logo

| S.No | Team Member<br>Name<br>(*) | WhatsApp<br>Contact<br>Number | Role in the Hackathon (#) | Designation(Employee/ UG/<br>PG Student / etc) | College / Organization<br>Name and Department<br>State & Country |
|------|----------------------------|-------------------------------|---------------------------|------------------------------------------------|------------------------------------------------------------------|
| 1.   | Nithin pranav              | 9500190447                    | SPOC                      | UG                                             | SATHYABAMA CSE,TN                                                |
| 2.   | RAM PRASANNA M             | 8098951293                    | MEMBER                    | UG                                             | SATHYABAMA CSE,TN                                                |
| 3.   | PESHAL L                   | 9043807881                    | MEMBER                    | UG                                             | SATHYABAMA CSE,TN                                                |
| 4.   | JEEVAN PRANAV              | 7010333104                    | MEMBER                    | UG                                             | SATHYABAMA CSE,TN                                                |
| 5.   | ROSHAN S                   | 9940375093                    | MEMBER                    | UG                                             | SATHYABAMA CSE,TN                                                |
| 6.   |                            |                               |                           |                                                |                                                                  |

(#) - Project Leader, Research Analyst, Developer, Tester / Debugger, Documentation Lead ,Presentation Lead, etc..

# Problem Statements

**Domain 1 – Open Challenges** - Identify and solve a real-world problem with an innovative, practical, and scalable solution aligned to any domain.

**Domain 2 – Business Solutions:** Improve user engagement, retention, and business value of an existing learning app through small, high-impact enhancements without redesigning it.

**Domain 3 – EdTech Solutions:** Optimize pre-launch user experience and engagement of an educational events platform to ensure strong adoption and retention..

**Domain 4 – IT Solutions:** Automate the conversion of raw DOCX manuscripts into publication-ready formats using AI with minimal manual effort.

**Domain 5 - Cyber Security & Forensic Science**

**Domain 6 - Healthcare & Allied**

# Instructions

## General Instructions

- This is a model submission template; teams may design their own version, but the first three slides are mandatory.

## For Round 2: Submission Requirements

- Teams must first submit their PPT, then present their detailed idea live on the ECLearnix/Gmeet platform. Presentations should clearly explain the solution and include answers to panel questions.
- All team members must actively participate in the online presentation.
- Only the **SPOC (Single Point of Contact) needs to** submit the Round 2 PPT (PPT saved as PDF format only). However, all team members must register on the ECLearnix portal to receive their certificates.  
File Name: **TeamName\_Round2.pdf**.

## Timelines

- Tentative Submission Window Starts: 08.06.2026 , Last date - 23.06.2026
- Tentative Presentation Dates: 27.06.2026

# Instructions

- Presentation duration: **12 minutes for presenting, plus 3 minutes for Q&A.**
- Ensure a stable internet connection, and test your microphone, camera, and screen sharing before presenting.
- Teams missing their allotted presentation slot will not be eligible for certificates & prizes.
- Only teams that present in Round 2 will receive a Presentation Certificate.
- Slide count limit: **Maximum 20 slides (excluding mandatory first three slides).**
- For technical support or queries, please contact: +91 9600888519
- Round 2 presentations will be evaluated for 25 marks total, with 5 marks allotted to each of the five criteria (outlined on the next slide).
- Bonus Marks = Social media sharing about your hackathon experience

## Note :

- “Presentations must be done using a laptop or computer;  
Mobile devices are not allowed.”

## Round 2 Presentation Rubrics

| S.No | Criteria                            | Description                                                                                                                          | Marks     |
|------|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 1    | Problem Understanding               | Clarity, relevance, depth of research, user pain points, domain alignment                                                            | 5         |
| 2    | Solution Design & Innovation        | Creativity, originality, practicality, feasibility, scalability                                                                      | 5         |
| 3    | Technical Implementation            | Architecture, workflow, accuracy, tools used, quality of demo/output                                                                 | 5         |
| 4    | Presentation Quality                | Slide clarity, communication, structure, time management, Q&A handling                                                               | 5         |
| 5    | Impact & Future Scope               | Real-world value, sustainability, user impact, growth/revenue potential, roadmap                                                     | 5         |
|      |                                     | <b>Total</b>                                                                                                                         | <b>25</b> |
| 6    | <b>BONUS: Social Media Sharing*</b> | <b>Sharing Hackathon experience on Social media platforms (Instagram, LinkedIn, YouTube, Facebook, Twitter (X), etc.) — Optional</b> | <b>5</b>  |

**“Sharing Your Experience on Social Media ✨  
You can post any videos, images, screenshots, shorts, or  
stories of your journey with us. Spread the word and let  
others be inspired!**

**Share your optional testimonial video about ECLearnix &  
Hackathon 4.0 to inspire and showcase our journey.**

**#ECLearnix #ECLearnixHackathon**

# Abstract & Domain Name

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## Domain 1 – Open Challenge Solutions

- **Problem Statement:** Insecure MQTT-based IoT networks vulnerable to unauthorized access, message tampering, and attacks.
- **Objective / Aim:** Design a low-cost, AI-powered security monitoring system using Raspberry Pi for continuous threat detection.
- **Proposed Solution / Approach:** Centralized Raspberry Pi broker employing AI anomaly detection to monitor IoT traffic and provide real-time web dashboard visualizations.
- **Expected Impact / Outcome:** Enhanced IoT security, low cost, scalability for smart environments.
- **File Format:** PPT only / **Submission:** via Hackathon Portal before the deadline





## MQTT Vulnerabilities

## User Pain Points



MQTT

### Widespread & Efficient

MQTT is widely used in IoT because it is lightweight and efficient.



### Communication Flaws

MQTT networks are vulnerable to unauthorized access and message tampering.



### Network Attack

Networks are vulnerable to flooding attacks and weak authentication.



### Lack of Oversight

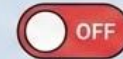
Many IoT environments lack real-time monitoring and threat detection.



Device compromise



Data theft



Service disruption



Lack of visibility into  
network threats

(Rubric: Problem Understanding)

# Objective of the Project

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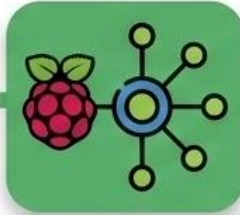
1. Monitor MQTT traffic in real time.



2. Detect vulnerabilities and suspicious activities.



3. Identify unauthorized devices and attacks.



4. Provide centralized monitoring using Raspberry Pi.



5. Visualize security alerts through a web dashboard.

- Explain
- AI-powered MQTT security monitoring.
- Raspberry Pi acts as MQTT broker and security analyzer.
- Dashboard provides live monitoring.
- Anomaly detection identifies cyber threats automatically.

## **Innovation:**

- Combines IoT, AI, and Cybersecurity.
- Low-cost deployment.
- Scalable for industrial use.
- Real-time detection capability.





## IoT Devices (Sensors/Actuators)

Generate sensor data, send  
input streams.



## MQTT Communication (Secure Protocol)

Plot or store centrally received  
data immediately.



## IoT Devices (Sensors/Actuators)

Generate sensor data,  
node, send input  
streams.

## MQTT Communication)

Publish and subscri-  
be messaging for  
data transport.

## Raspberry Pi (Compute Node)

Acts as the central  
hub, data  
aggregator, and  
initial processor.

## AI Analysis (Machine Learning Models)

Processes data for  
anomaly detection  
& threat analysis.

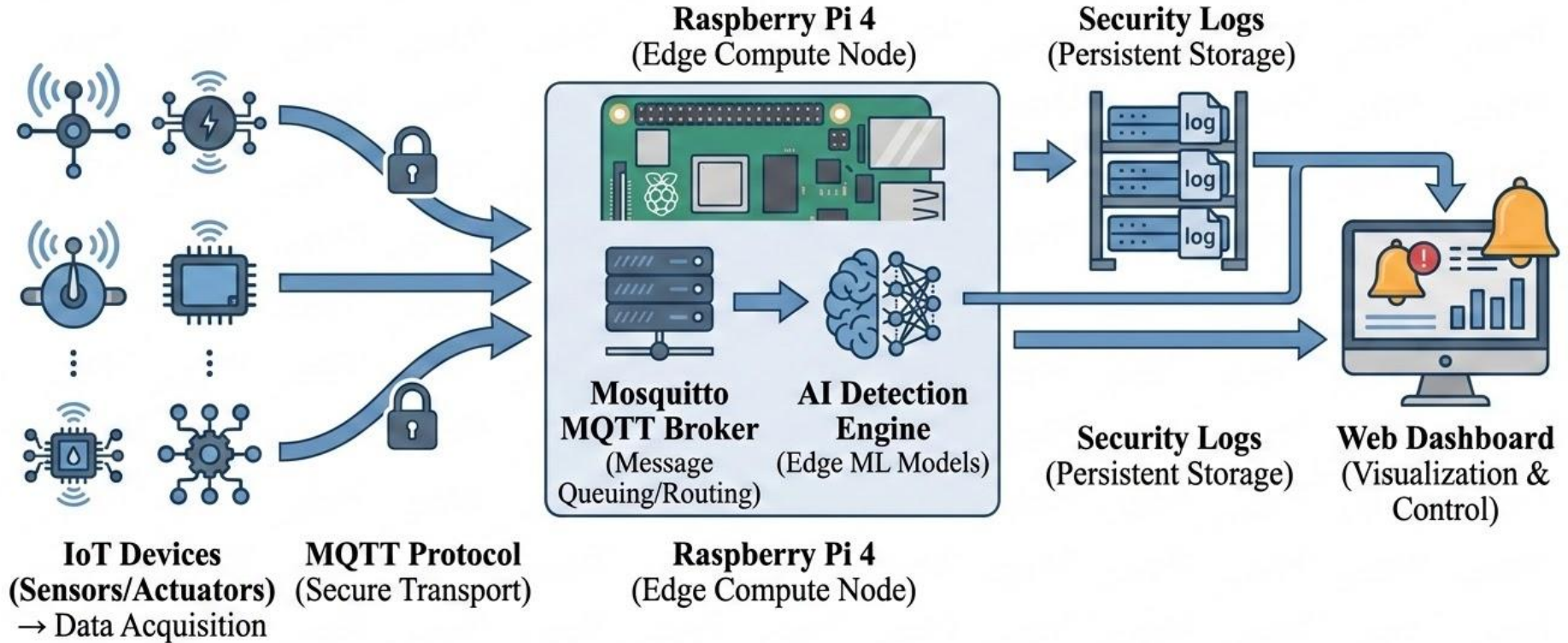
## Threat Detection (Real-time Analytics)

Identify security  
breaches or  
anomalies  
immediately.

## Dashboard Alerts (Visualization & Notification)

Display real-time  
security status and  
notify operators.

# Architecture Diagram





## Real-Time MQTT Monitoring

Continuous, low-latency tracking of all connected sensor and actuator traffic.



## AI-Based Anomaly Detection

Leveraging machine learning models to identify unusual network behavior.



## Unauthorized Access Detection

Monitoring logins and connection attempts to prevent unauthorized device access.



## Message Tampering Detection

Identifying modification, interception, or alteration of interception, or alteration of transmitted MQTT messages.



## Dashboard Alert System

Immediate visual and audible notifications for critical security events.



## Centralized Security Monitoring

Unified oversight and analysis of security logs from all distributed edge nodes.





**Raspberry  
Pi**

**Single-board Computer**

Main computing unit for data processing and control.



**MQTT**

**Messaging Protocol**

Lightweight, publish-subscribe protocol for efficient IoT



**Mosquitto  
Broker**

**Message Broker**

Standard-compliant MQTT broker managing all client connections.



**Python**

**Programming Language**

Primary language for system logic, scripting, and data analysis.



**Flask/  
FastAPI**

**Web Framework**

Tools for building a responsive web interface and APIs.

HTML



CSS



JS



**Frontend Technologies**

Powering the user dashboard and UI presentation.



**AI/ML**

**Intelligent Analysis**

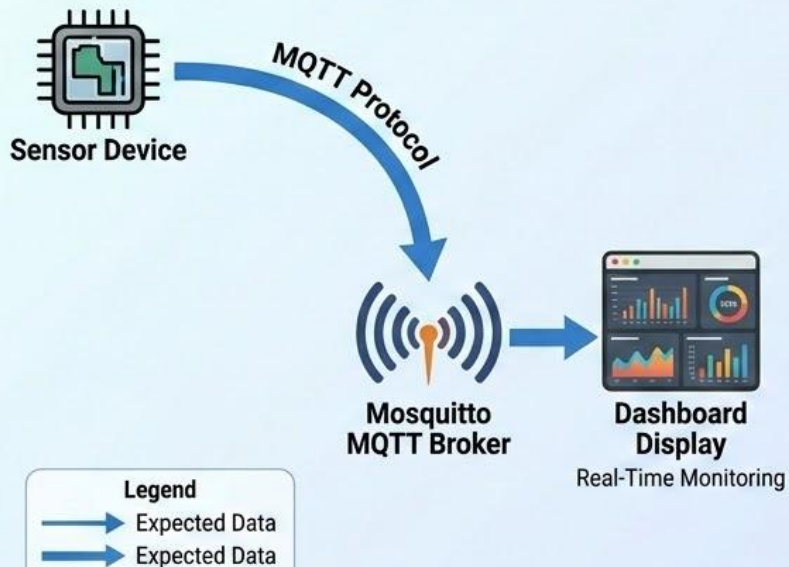
Implementing machine learning algorithms for advanced insights.

**WIRESHARK**

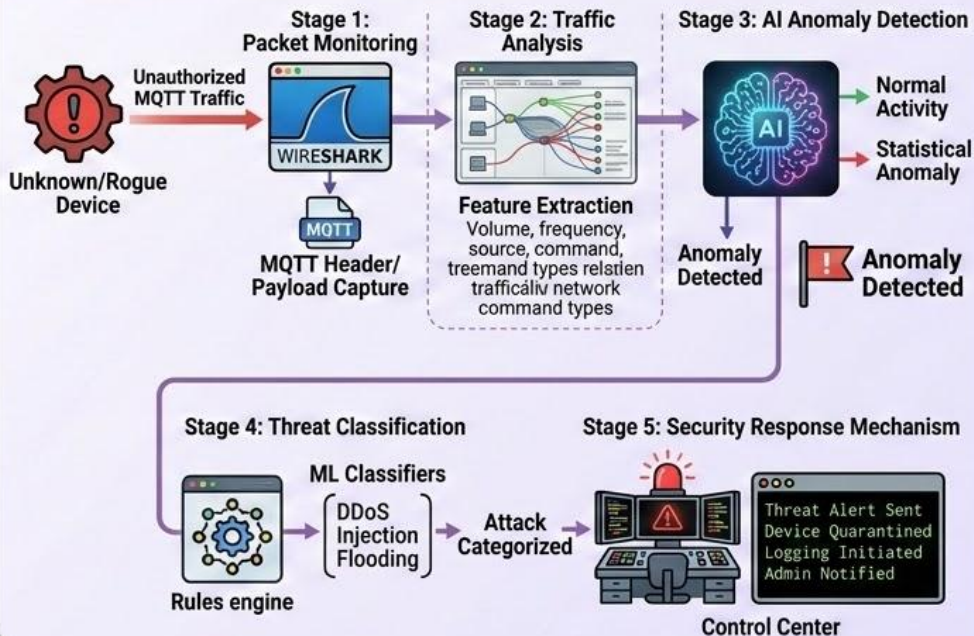
**Network Analysis Tool**

Capturing and analyzing network traffic for monitoring and debugging.

## Normal IoT Data Flow (Baseline)



## AI-Powered Threat Detection Flow (Anomaly)





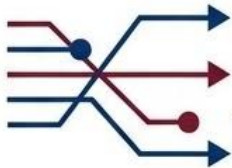
- Project/demo Videos
- GitHub: <https://github.com/rahul180306/neuroguard.g>

|                  | Traditional Monitoring Systems | CODERX AI Solution (MQTT-Focused)  |
|------------------|--------------------------------|------------------------------------|
| AI Detection     | X                              | ✓ AI-driven detection              |
| Monitoring Speed | Periodic / Reactive            | Real-time monitoring               |
| Protocol Focus   | General                        | 🔒 MQTT-focused security            |
| Deployment Cost  | High-cost infrastructure       | 🍷 Raspberry Pi low-cost deployment |
| Architecture     | Monolithic / Fixed             | ⬆️⬆️ Scalable architecture         |

## Innovation Highlights



- 🧠 Advanced AI Detection Network
- 🔔 Instant Real-time Alerts
- 🔒 Secure MQTT Protocol
- 💰 Cost-Effective Edge Computing



## 1. Smart Homes

### A. Improved Security

- Secure home systems

### B. Reduced Attack Risk

- Prevent hacking of smart locks/ and dr cameras

### C. Real-Time Visibility

- Constant device status

### D. Low-Cost Implementation

- Affordable gateway deployment



## 2. Healthcare IoT

- HIPAA compliance

- Protect patient devices (pacemakers, monitors)

- Instantaneous health data monitoring

- Efficient hospital network protection



## 3. Industrial IoT

- Supply chain security

- Block DDoS/ ransomware on factory floor

- Production line monitoring

- Cost-effective industrial gateway security



## 4. Smart Agriculture

- Drone and sensor security

- Protect remote infrastructure from tampering

- Real-time crop/ livestock data

- Low-cost sensor protection on expansive farms



## 5. Smart Cities

- Traffic light and grid security

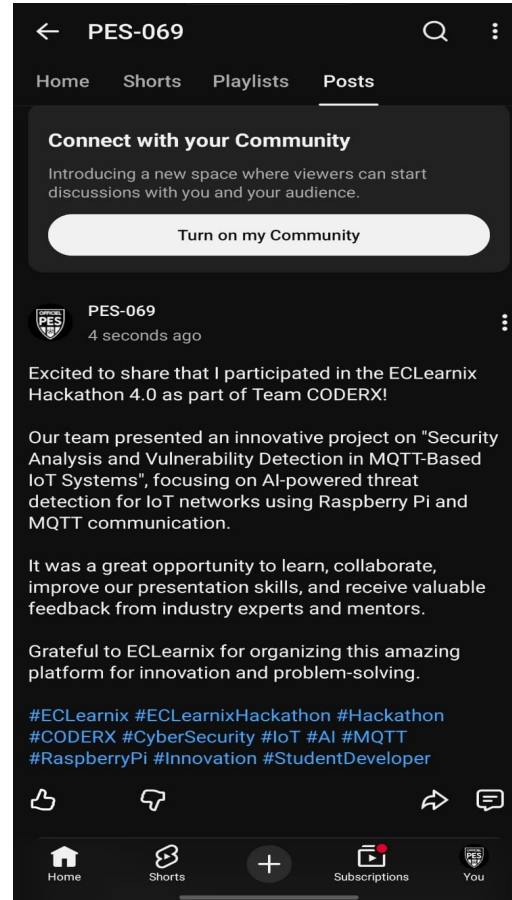
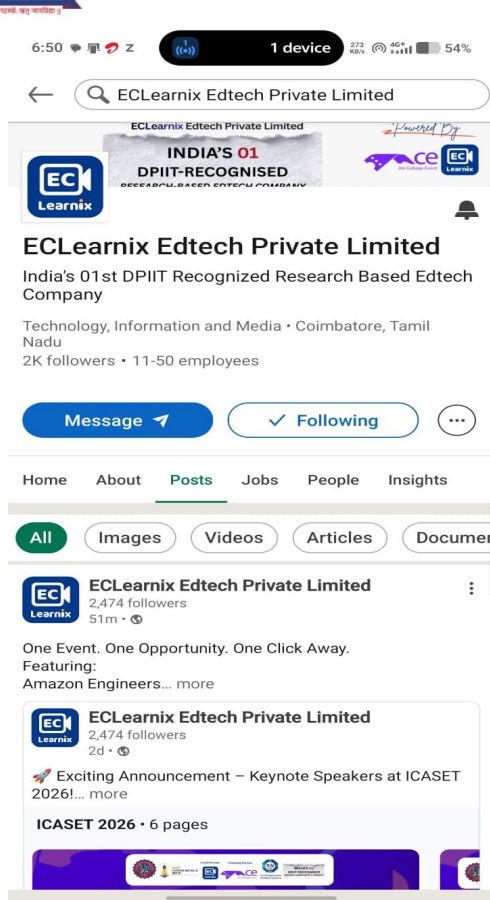
- Secure surveillance networks

- City-wide infrastructure monitoring

- Affordable city-wide network security integration

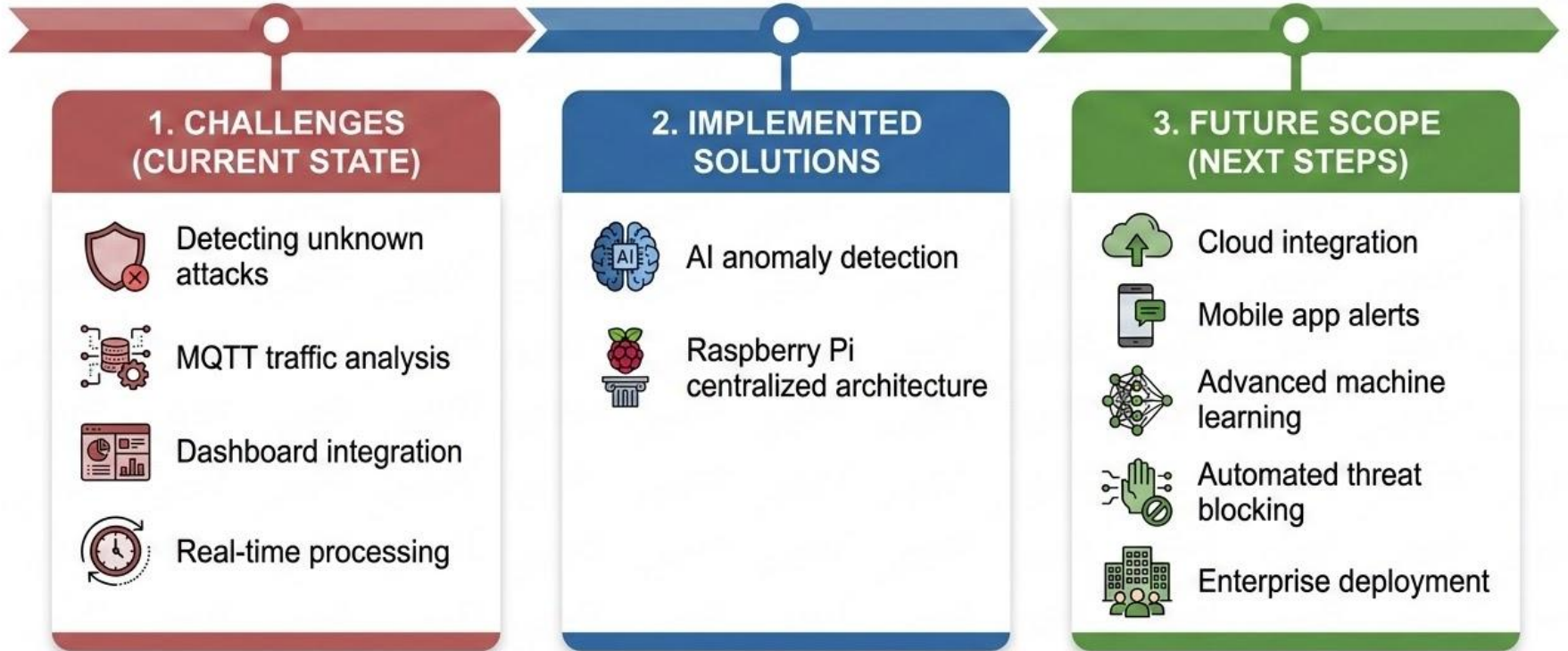
# Bonus Slide (Optional)

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(Rubric: Presentation + Technical)





# Conclusion & Thank You

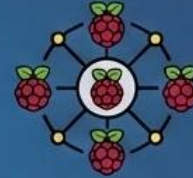
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MQTT security  
problem  
addressed



AI-powered  
monitoring  
solution



Raspberry Pi  
centralized  
architecture



Real-time threat  
detection



Practical and  
scalable  
deployment

## Building Safer IoT Networks Through AI-Driven Cybersecurity.

TEAM CODERX